

Exercise - Week 10: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Equivalence Relations, Partial Order Relations, Definition of Groups

Student Name:

Student number: |_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

(1) Define a relation ρ on $\mathbb{Z}_{\geq 1}$ by $a\rho b$ if

$$\gcd(a, 144) = \gcd(b, 144).$$

(a) **(3 marks)** Prove that ρ is an equivalence relation.

(b) **(2 marks)** Determine the equivalence classes. How many equivalence classes are there? Justify your answer.

- (2) (**3 marks**) Let S be the set of binary strings of length 6. The weight of a binary string is the number of ones it contains, for example $\text{weight}(110110) = 4$. Define a relation ρ on S by

$$s \rho t \quad \text{if and only if} \quad \text{weight}(s) \leq \text{weight}(t).$$

Is ρ a partial order? Justify your answer.

- (3) (**2 marks**) Determine whether $\mathbb{Z}_{10} - \{0\}$ is a group under multiplication.